

Introduction

You are tasked with processing data related to movies, actors, and directors, stored in RDDs. The formats of the RDDs are as follows:

- **Movies** RDD: Each element of **Movies** is a tuple $(id_movie, title, language)$.
Example: (1, "The Matrix", "English"), (2, "Inception", "English")
- **Actors** RDD: Each element of **Actors** is a tuple $(id_movie, actor)$, indicating the actors who starred in a movie. Example: (1, "Keanu Reeves"), (1, "Laurence Fishburne")
- **Directors** RDD: Each element of **Directors** is a tuple $(id_movie, director, nationality)$, indicating the director of a movie and its nationality.
Example: (1, "Lana Wachowski", "USA"), (2, "Christopher Nolan", "England")

1 Important instructions

1. You should write the code to create the given RDD in each question.
2. You can of course use a RDD build in a question n to answer another question m (with $m > n$), even if you did not answer question n .
3. If you build intermediate RDD to answer a question, explain what is this RDD and its format.
4. The Spark functions you can use to solve the exercises are: **map**, **flatMap**, **reduceByKey**, **filter**, **union** and **groupByKey**. If A is a RDD of pairs $(key, value)$, then $A.groupByKey()$ is a RDD of pairs $(key, list_of_value)$ where the values with the same key are grouped together in the same list. Other functions are not allowed.

2 Questions

Question 1 : Create a **FrenchTitles** RDD containing only the titles of the movies where the language is French.

Question 2 : Create a **ActorStats** RDD containing tuples $(name, nba)$ where nba is the number of movies where $name$ is an actor.

Question 3 : Create a **Stats** RDD containing tuples $(name, (nba, nbd))$ where nba is the number of movies where $name$ is an actor, and nbd is the number of movies where $name$ is a director (one of these values can of course be 0).

Question 4 : Create a **ActorsList** RDD containing tuples $(id_movie, list_of_actors)$ where the list contain all actors of a given movie.

Question 5 : Create a **Multiple** RDD containing tuples $(actor, director)$ where $director$ directed $actor$ on multiple movies (at least 2).